BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (MCQ's)

Note: Q.1: Choose the correct answer for the the given option:

Q1. (a3x + a3y)/(a(x-y)) = _____

(a) ax2+ ay2
(b) x2 + y2
(c) a(x2- xy + y2)
(d) a (x2 +xy + y2)

Q2. _____ is th e equidistainct for all these sides of a triangle.

(a) incentre
(b) Circumcentre
(c) Orthocentre
(d) Cntroid

Q3. Base in Natural logearithm is _____

(a) 10
(b) e
(c) pi
(d) 5

Q4. How many maximum rootss of a Quadratic equation are:

(a) 2
(b) 3
(c) 5
(d) 4

Q5. PQR - m angle P = 65 degrees and m angle Q = 57 degrees then _____ will be teh shortest side.

(a) PQ
(b) QR
(c) RP
(d) None of these

Q6. Factors of a2 + 2ab + b2 - c2 is:

(a) (a-b-c)(a-b-c)
(b) (a+b+c)(a-b-c)
(c) (a+b+c)(a+b-c)
(d) (a+b+c)(a-b-c)

Q7. The line x = 6 is parallel to :

(a) x-axis
(b) y-axis
(c) both axis
(d) None of these

Q8. (a + b)(a - b)(a2 - b2) = _____.

(a) (a2 - b2)2
(b) a4 - b4
(c) (a2 + b2)
(d) (a4 + b4)

Q9. Solution se of sqrt(y4) = 10 is :

(a) {100}
(b) {10}
(c) {-10}
(d) {-10,10}

Q10. The bisector of the angles of a traingle are:

(a) Concrurrent
(b) Collinear
(c) Unequal
(d) Do not inersect

Q11. Standard form of Quadratic equation is:

(a) ax2 +bx + c 0, a != 0
(b) ax2 + c = 0, a != 0
(c) ax2 + bx = 0 a != 0
(d) ax3 + bx2 + c = 0, a != 0

Q12. sqrt(y)+4 + 2 = 8 is a :

(a) linear equation
(b) Radical equation
(c) cubic equation
(d) Quadratic equation

Q13. A triangle havign all the three sides congruent is called _____ triangle.

(a) Scalene
(b) Right-angled
(c) equilateral
(d) Isosceles

Q14. HCF of x2 - [illegible in source scan] and x2 - 4xy + 4y2 is:

(a) x - 4y
(b) x2 + 2xy +- y2
(c) x + 2y
(d) x - 2

Q15. QUO' = _____

(a) R
(b) E
(c) N
(d) W

Q16. Scientifc notation of 0.00789 is _____.

(a) 7.89 x 103
(b) 7.89 x 10
(c) 0.789x102
(d) 78.9 x 104

Q17. The value of 't' for which 49x - 42xy + 1 will be a pretect square:

(a) 9
(b) 9y2
(c) 3y2
(d) 3

Q18. (5+sqrt5)(5 - sqrt15) is equal to:

(a) 10
(b) 15
(c) 25
(d) 30

Q19. E4 is closed under:

(a) Division
(b) Multiplication
(c) Substraction
(d) All of these

Q20. (5i).(-2i) = _____

(a) -10
(b) 10
(c) -10i
(d) 10i

Q21. log(a/bcd) = _____

(a) loga -log b + logc + logd
(b) loga - logb + logc - logd
(c) loga - logb - logc - logd
(d) loga - log - logc + logd

Q22. The degree of the 3xy + 5y4 - 10 is

(a) 4
(b) 5
(c) 6
(d) 10

Q23. The _____ of a triangle cut each other in the ratio 2:1 :

(a) Altitude
(b) Median
(c) angle bisector
(d) Right bisector

Q24. If the sum of measures of angle A and angle C of a parallelogram ABCD is 140 degrees then angle D will be of _____.

(a) 70 degrees
(b) 10 degrees
(c) 140 degrees
(d) 220 degrees

Q25. ((x1 + x2)/2, (y1 + y2)/2) is called:

(a) Mid point formula
(b) Distance formula
(c) Division formula
(d) Ratio formola

Q26. _____ are the length of sides of a right angle triangle.

(a) 3, 4, and 5
(b) 4, 3 and 5
(c) 2, 4,and 6
(d) None of these

Q27. A point equidistant form the end points of a line segment is on the _____ of it.

(a) Perpendicular
(b) Right bisector
(c) Centre
(d) Bisector

Q28. The point (-3, -4) is located in:

(a) 1st Quadrant
(b) 2nd Quadrant
(c) 3rd Quadrant
(d) 4th Quadrant

Q29. If log10 x = 4, then x = _____

(a) 500
(b) 100
(c) 1000
(d) 10000

Q30. In siplified form 1/(x+y) + y/(x2 + y2) = _____

(a) (y+1)/(x2+y2)
(b) x/(x2+y2)
(c) y/(x2+y2)
(d) (y-1)/(x2-y2)

Section-B

> Note: Attempt any NINE of the following questions:

Q2. Simplify: (i) 2^4 5^3 / 10^3 (ii) [illegible in source scan]

Q3. If Log 3 = 0.4711 Log 2 = 0.3010 Log 5 = 0.6990 then find the value of Log 30

Q4. Ifab = 5 andab=7, find the value ofa3b3. Find the factors of the following

Q5. (i) x2- 1 + 1x2/4x2 (ii) (2x + x)2 - (2x -x)2

Q6. Find the LCM of x2 +11x + 28 and x2 + x - 12

Q7. Simplify (ii) sqrt(x) - 8 = 1 (ii) |5x - 12| = 7

Q8. Solve 9x2 = 12x - 49 by quadratic formula

Q9. solve sqrt(x+2) + sqrt(x+7) = sqrt(6x+13)

Q10. Find the value of x and y when x+ yt = -5 + 5i

Q11. x = 3 - 2sqrt(2) find x2 + 1/x2-

Q12. If the polynomial 4x3 - 7x2 + 6x - 3k Is exactly divisible by (x + [illegible in source scan]) find the value of K

Q13. Find the square root by division method [illegible in source scan] 2x - 3x2 + 2x + 1

Q14. Find tne number of digits in 45

SECTION - C

> Note: ATT9AP: ANT 1-015E1010F THE fOLLOwsit3 QUESTIONS.

Q15. If two angles of a triangle are congruent then opposite sides will also be congruent

Q16. Define with Diagram (i) Right Bisector of fine segment (ii) Equilateral Triangle (iii) Congruent Triangles

Q17. If opposite sides of a quadrilateral are congruent and parallel then quadrilateral will be parllogram.

Q18. Draw triangle ABC. if m angle A = 65 degrees m.AC = mA8 = 5 lcm

Q19. Find am central point of a circle when the end points of diameter are A[illegible in source scan]5 and B(3-4).